

CENG381 Stochastic Processes

Product-Form Martingales and Jensen's Inequality — Practice 1 (Answer Key)

Q1 Solution

a) Product-form martingale proof:

$$\begin{aligned}Z_n &= Z_{\{n-1\}} * X_n \\E[Z_n \mid Z_{\{n-1\}}, \dots, Z_1] &= E[Z_{\{n-1\}} * X_n \mid Z_{\{n-1\}}, \dots, Z_1] \\&= Z_{\{n-1\}} * E[X_n \mid Z_{\{n-1\}}, \dots, Z_1] \\&= Z_{\{n-1\}} * E[X_n] \quad (X_n \text{ independent of past}) \\&= Z_{\{n-1\}} * 1 = Z_{\{n-1\}} \quad \checkmark\end{aligned}$$

b) Example $X_n = 2$ w.p. $1/2$, 0 w.p. $1/2$:

$$\begin{aligned}E[X_n] &= 2*(1/2) + 0*(1/2) = 1 \quad \checkmark \\Z_n &= X_1 * \dots * X_n \\&= 2^n \quad \text{if all } X_i = 2 \quad (\text{prob } (1/2)^n) \\&= 0 \quad \text{if any } X_i = 0 \quad (\text{prob } 1 - (1/2)^n) \\E[Z_n] &= 2^n * (1/2)^n + 0 = 1 \quad \text{for all } n \\P(Z_n > 0) &= (1/2)^n \rightarrow 0 \quad \text{as } n \rightarrow \infty \\Z_n &\rightarrow 0 \quad \text{almost surely} \quad (\text{eventually some } X_i = 0, \text{ making product } 0) \\E[Z_{\infty}] &= E[0] = 0 \neq \lim E[Z_n] = 1\end{aligned}$$

c) Why this does not contradict the martingale definition:

The martingale definition only requires $E[Z_n \mid \text{past}] = Z_{\{n-1\}}$ for each finite n — it says nothing about what happens at $n = \infty$. The martingale convergence theorem guarantees $Z_n \rightarrow Z_\infty$ a.s. under certain conditions, but does NOT guarantee $E[Z_\infty] = \lim E[Z_n]$ unless $\{Z_n\}$ is also uniformly integrable.

Uniform integrability (UI) fails here: the distribution of Z_n is increasingly concentrated on the single very large value 2^n (with tiny probability 2^{-n}), making the tails unbounded. UI would require $\sup_n E[Z_n \cdot 1_{\{Z_n > M\}}] \rightarrow 0$ as $M \rightarrow \infty$, which fails.

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Q2 Solution

a) Jensen for $g(x) = x^2$:

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(i)  $g''(x) = 2 > 0 \Rightarrow g$  is convex ✓  
  
(ii) Jensen:  $E[X^2] \geq (E[X])^2$   
 $\Rightarrow E[X^2] - (E[X])^2 \geq 0$   
 $\Rightarrow \text{Var}(X) \geq 0$  (variance is always non-negative) ✓
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b) Jensen for $g(x) = e^x, X \sim \text{Normal}(\mu, \sigma^2)$:

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Jensen:  $E[e^X] \geq e^{E[X]} = e^\mu$   
  
For  $X \sim N(\mu, \sigma^2)$ :  $E[e^X] = e^{\mu + \sigma^2/2}$   
  
Check:  $e^{\mu + \sigma^2/2} \geq e^\mu$   
iff  $\mu + \sigma^2/2 \geq \mu$   
iff  $\sigma^2/2 \geq 0$  ✓ (always true)  
  
Equality holds iff  $\sigma^2 = 0$  ( $X$  is deterministic =  $\mu$ ).
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c) Jensen for $g(x) = -\log(x)$; AM-GM inequality:

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 $g''(x) = 1/x^2 > 0 \Rightarrow g$  is convex for  $x > 0$  ✓  
  
 $X = a_i$  w.p.  $1/n$  for  $i=1, \dots, n$   
 $E[X] = (a_1 + \dots + a_n)/n = \text{AM}$   
 $E[g(X)] = (1/n) * (-\log a_1 - \dots - \log a_n) = -\log(\text{GM})$   
where  $\text{GM} = (a_1 * \dots * a_n)^{1/n}$   
  
Jensen:  $E[g(X)] \geq g(E[X])$   
 $-\log(\text{GM}) \geq -\log(\text{AM})$   
 $\log(\text{GM}) \leq \log(\text{AM})$   
 $\text{GM} \leq \text{AM}$  ✓  
  
 $(a_1 * a_2 * \dots * a_n)^{1/n} \leq (a_1 + a_2 + \dots + a_n)/n$ 
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Q3 Solution

a) Submartingale via Jensen:

$$\begin{aligned} & E[h(Z_n) \mid Z_{\{n-1\}}, \dots, Z_1] \\ & \geq h(E[Z_n \mid Z_{\{n-1\}}, \dots, Z_1]) \quad [\text{Jensen, since } h \text{ convex}] \\ & = h(Z_{\{n-1\}}) \quad [\text{martingale condition}] \end{aligned}$$

So $E[h(Z_n) \mid \text{past}] \geq h(Z_{\{n-1\}})$, i.e., $\{h(Z_n)\}$ is a sub-martingale. ✓

b) $\{S_n^2\}$ is a submartingale:

$$\begin{aligned} \text{(i)} \quad & E[S_n^2 \mid S_{\{n-1\}}] \\ & = E[(S_{\{n-1\}} + X_n)^2 \mid S_{\{n-1\}}] \\ & = S_{\{n-1\}}^2 + 2S_{\{n-1\}}E[X_n] + E[X_n^2] \\ & = S_{\{n-1\}}^2 + 0 + 1 \\ & = S_{\{n-1\}}^2 + 1 \geq S_{\{n-1\}}^2 \quad \checkmark \quad (\text{sub-martingale}) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad & \{S_n^2 - n\} \text{ is a martingale} \Rightarrow E[S_n^2 - n] = E[S_0^2 - 0] = 0 \\ & E[S_n^2] = n \end{aligned}$$

c) $\{|S_n|\}$ is a submartingale; CLT check:

$h(x) = |x|$ is convex (V-shape; $h'(x) = 0$ for $x \neq 0$ and the kink at 0 satisfies the convexity condition).

By Jensen applied to the martingale $\{S_n\}$:

$$E[|S_n| \mid S_{\{n-1\}}, \dots] \geq |E[S_n \mid S_{\{n-1\}}, \dots]| = |S_{\{n-1\}}|$$

So $\{|S_n|\}$ is a sub-martingale. ✓

$E[|S_n|]$ is increasing in n (as required for a sub-martingale).

CLT check: $E[|S_n|] \sim \sqrt{2n/\pi}$ which is increasing in n ✓

(The sequence grows like $\sqrt{n}$, consistent with the walk spreading out.)